

Soil organic carbon dynamics under no-tillage and conventional tillage in a semi-arid durum wheat system in Northwest Algeria

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Abstract: Soil organic carbon is crucial for maintaining soil fertility, water retention, and resilience to climate change, particularly in semi-arid regions. This study explores the effects of no-till and conventional tillage systems on soil organic carbon dynamics in durum wheat (*Triticum durum* Desf.) crops in northwestern Algeria. Field experiments were conducted over two growing seasons (2022–2024) to assess temporal and spatial changes in organic carbon content and its relationship with other soil parameters, and to provide a preliminary ten-year projection of organic carbon levels. Results showed numerical trends suggesting greater carbon stability and accumulation under no-till, while conventional tillage exhibited higher temporal variability and a tendency to decline. Principal component analysis suggested an important role for organic carbon in structuring soil quality patterns under no-till. Linear projections indicated that no-till systems may reach approximately 2.3% by 2034, whereas conventional tillage is projected to remain near 1.2%. However, these projections are heuristic and based on a short-term dataset. Both systems remain below their theoretical soil organic carbon saturation levels, suggesting potential for further sequestration under appropriate management. Overall, these findings provide exploratory evidence of the potential of conservation agriculture to maintain or enhance soil carbon in semi-arid Mediterranean regions and highlight the need for long-term monitoring and integrated management strategies.

Keywords: carbon dynamics, conservation agriculture practices, direct seeding, Mediterranean agroecosystems, soil carbon trends, soil fertility

INTRODUCTION

The soil, a dynamic interface between the atmosphere and the lithosphere, plays a central role in the functioning of terrestrial ecosystems. It provides essential ecosystem services, including the recycling of organic matter (OM), nutrient storage, pollutant degradation, regulation of the hydrological cycle, and carbon sequestration (Desrousseaux *et al.*, 2019). In particular, soil organic carbon (SOC) is a key indicator of soil health and its ability to mitigate climate change (Lal, 2004). Since the pre-industrial era, the increase in anthropogenic greenhouse gas emissions (CO₂, CH₄, N₂O) has led to unprecedented global warming (Pachauri *et al.*, 2014). In this context, agricultural soils, which represent the largest terrestrial reservoir of organic carbon (OC) (Scharlemann *et al.*, 2014), can play a crucial role by acting

either as a sink or a source of atmospheric CO₂, depending on management practices (Dignac *et al.*, 2017; Chenu *et al.*, 2020).

Cropping systems, particularly in semi-arid areas, are strongly influenced by tillage techniques, which alter SOC dynamics. Two main methods are compared: conventional tillage (TC), which involves ploughing, and no-till farming, also known as direct seeding (SD), which does not involve ploughing. Ploughing, although it temporarily improves aeration and residue incorporation, accelerates the mineralisation of OM, leading to a decrease in SOC stocks (Six *et al.*, 2002). Conversely, SD, by limiting soil disturbance, promotes carbon accumulation in surface horizons (López-Fando and Pardo, 2011) and improves structural stability, water retention, and soil biodiversity (Kassam, Friedrich and Derpsch, 2019).

Recent studies confirm that Algeria is experiencing increased climate stress, with a trend toward rising average temperatures and declining precipitation over the past few decades (Benassi *et al.*, 2020; Marco *et al.*, 2022). This trend exacerbates the challenges associated with soil degradation and declining soil fertility, particularly in traditional cereal farming systems (Moussadek *et al.*, 2024). In this context, the evaluation of adaptive agricultural practices, such as SD, is crucial to mitigate the effects of aridity while preserving soil OM (Boudiar *et al.*, 2022). However, quantitative data on the effectiveness of SD in sequestering SOC in semi-arid cereal systems in Algeria remain limited, particularly over multi-year timescales.

Several studies in arid and semi-arid areas have demonstrated the agronomic and environmental benefits of SD. In Algeria, Abdellaoui *et al.* (2011) observed improved growth of durum wheat under SD, attributed to an improvement in the physical and chemical properties of the soil. Similarly, Sturny *et al.* (2007) emphasised that SD ensures long-term fertilisation while reducing production costs. Mekhlouf *et al.* (2011) also highlighted its positive impact on soil moisture, bulk density, and porosity. A recent study by Boudiar *et al.* (2022) confirmed that no-till systems significantly increase SOC stocks compared to conventional tillage, while reducing greenhouse gas emissions linked to OM mineralisation and the use of agricultural machinery. However, Moussadek *et al.* (2024) noted that cereal yields often remain limited in conservation systems, particularly due to soil and climate constraints, as well as the incomplete adoption of associated techniques.

Although numerous studies have assessed the impact of farming techniques on SOC in temperate regions, few have focused on semi-arid areas, where soils are particularly vulnerable to degradation and desertification. Our study aims to: (i) quantify the effect of SD on SOC sequestration over two years of monitoring, compared with TC, in a durum wheat cropping system in the Sidi Bel Abbes region (northwestern Algeria); (ii) evaluate SOC dynamics during the phenological stages of the crop to identify critical periods of carbon loss or stabilisation; (iii) project, through linear modelling, the ten-year evolution of SOC until 2034, taking into account critical fertility thresholds and theoretical saturation points specific to each system.

The central hypothesis is that SD, by limiting physical disturbance of the soil and promoting the return of residues, may contribute to increasing OC stocks compared to ploughing, with greater long-term sequestration potential.

Unlike previous studies, this study combines a detailed temporal approach (biannual monitoring) with an integrated analysis of soil agronomic properties to propose practical recommendations for resilient agriculture in semi-arid areas. Our results will contribute to the optimisation of sustainable soil management policies aligned with food security objectives.

MATERIALS AND METHODS

STUDY AREA CHARACTERISTICS

The Technical Institute for Field Crops (Fr.: Institut Technique des Grandes Cultures – ITGC) is located 4 km east of Sidi Bel Abbes at an altitude of 491 m (35°10'26" N, 0°40'19" W). The region has a semi-arid climate, classified in the lower category of

the Emberger system (Meterfi, 2014), with high summer temperatures often exceeding 35°C and low, irregular annual rainfall below 400 mm (Faraoun, 2014). The De Martonne aridity index of 9.65 further confirms conditions at the threshold between semi-arid and arid (Yahiaoui-Chebre *et al.*, 2022), leading to high potential evapotranspiration and limited water availability for rain-fed crops, particularly cereals (Meddi, Toumi and Assani, 2016).

The soils are classified as Calcisols, with a loamy texture (silt content 35–40%), a limestone crust at 20–45 cm depth, and high total calcium carbonate content (31–35%) (Faraoun, 2014).

SITE DESCRIPTION AND SAMPLING STRATEGY

The study was carried out under semi-arid conditions during two cropping seasons (2021/2022 and 2023/2024). In the 2022/2023 intermediate season, chickpea was cultivated in the form of worked fallow across the entire experimental field. At the time of sampling, the no-till system had been practised continuously at the site for seven years (since 2015). The experimental design was identical in both seasons. Two adjacent plots were established and each plot was subdivided into three blocks (replicates) of equal size (300 m²). The first plot was managed under no-till (SD) with three replicates (one block each), whereas the second plot was managed under conventional tillage (TC) with three replicates (Fig. 1). In both management systems, durum wheat (*Triticum durum* Desf.) was cultivated following local agronomic practices (Tab. 1).

Soil sampling was performed in each season at five key phenological stages of the crop. Samples were collected from the 0–30 cm soil layer in the six experimental blocks (three SD and three TC). Within each block, seven soil cores were randomly taken using a soil auger and thoroughly mixed to obtain one representative composite sample per block. This composite sampling strategy was adopted to reduce small-scale spatial variability and improve the representativeness of the measurements.

The sampling stages were defined as follows: P1: before sowing (06 Oct 2021 for 2021/2022; 06 Nov 2023 for 2023/2024); P2: germination/emergence (08 Jan 2022; 31 Jan 2024); P3: tillering (22 Feb 2022; 21 Feb 2024); P4: heading stage (29 Mar 2022; 16 Apr 2024); P5: physiological maturity (16 May 2022; 13 May 2024).

The initial sampling (P1) was conducted prior to tillage operations in TC and prior to sowing in SD in order to establish the baseline physicochemical status of the soil. Subsequent samplings (P2–P5) were used to monitor the temporal dynamics of SOC and associated soil properties throughout the wheat growing cycle.

Durum wheat (*Triticum durum* Desf.) was cultivated in all monitored plots according to the technical procedures outlined in Table 1.

PHYSICAL AND CHEMICAL ANALYSES

Particle size distribution was determined using the Robinson method (Gee and Or, 2002). This method involves dispersing the samples in sodium hexametaphosphate after removing cements and OM by oxidation with hydrogen peroxide. The fractions were separated by pipetting at different times and depths according to

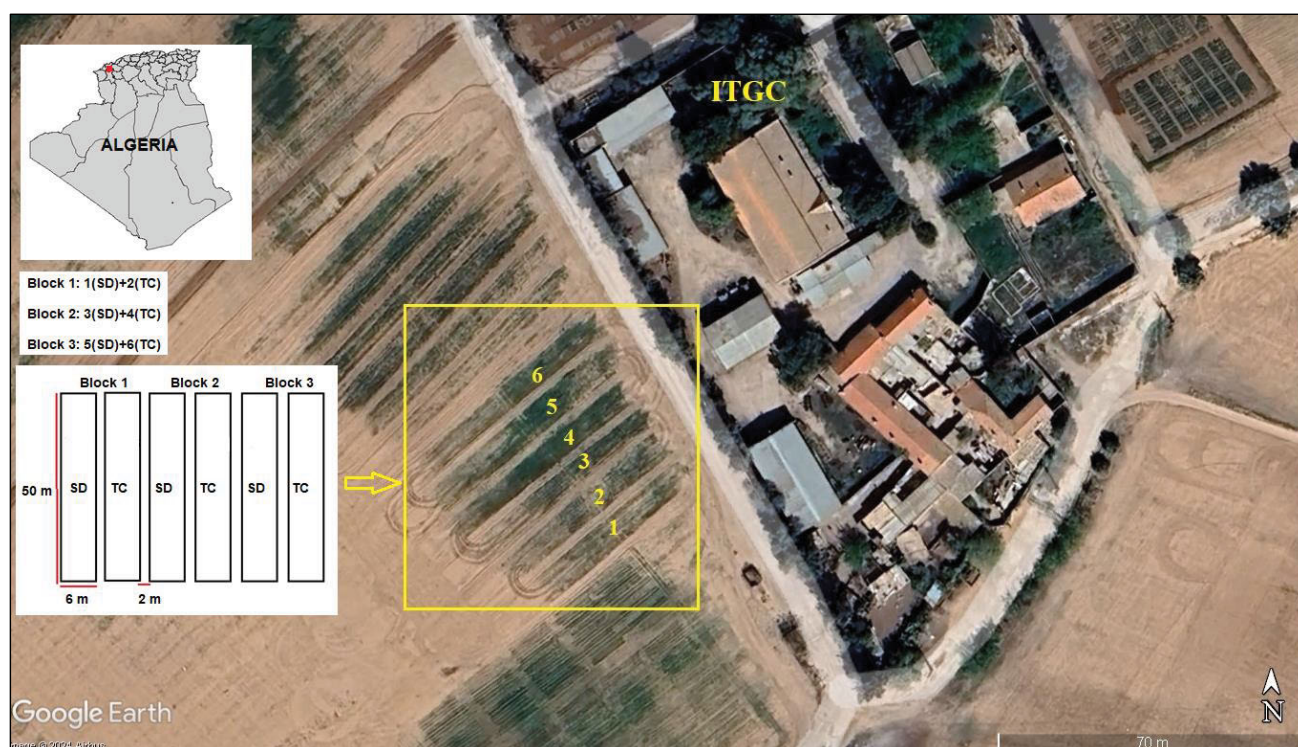


Fig. 1. Location and experimental layout of the trial plots at the Technical Institute of Field Crops (ITGC) in Sidi Bel Abbes, showing the distribution of blocks according to the two tillage systems: no-till (SD) and conventional tillage (TC); source: own elaboration based on Google Earth

Table 1. Summary of field operations and crop management for durum wheat (*Triticum durum* Desf.) under no-till (SD) and conventional tillage (TC) systems during the 2021–2022 and 2023–2024 cropping seasons

Operation	System	Date	Material/Product	Rate/Parameter	Objective/Remarks
Cycle 2021–2022					
Previous crop	SD/TC	2020–2021	chickpea	–	cereal–legume rotation
Deep tillage	TC	01 Nov 2021	three-share plough	–	primary soil preparation
Basal fertilisation	SD/TC	10 Nov 2021	matrix NP (7–17%)	150 kg·ha ⁻¹	basal NP supply
Sowing	TC	07 Dec 2021	conventional seeder	150 kg·ha ⁻¹ (TGW: 37 g)	density: 300 plants per m ²
Sowing	SD	07 Dec 2021	John Shearer no-till seeder	150 kg·ha ⁻¹ (TGW: 37 g)	density: 300 plants per m ²
Chemical weeding 1	SD/TC	13 Nov 2021	glyphosate	4.5 dm ³ ·ha ⁻¹	non-selective; applied pre-planting to avoid crop damage
Chemical weeding 2	SD/TC	23 Feb 2022	Floramix	300 g·ha ⁻¹	selective post-emergence with safener to protect the crop
Nitrogen fertilisation 1	SD/TC	03 Mar 2022	urea (46% N)	100 kg·ha ⁻¹	first supplementary application
Nitrogen fertilisation 2	SD/TC	05 Apr 2022	urea (46% N)	100 kg·ha ⁻¹	second supplementary application
Irrigation 1	SD/TC	30 Jan 2022	sprinkler system	30 mm (6 h)	winter irrigation
Irrigation 2	SD/TC	16 Feb 2022	sprinkler system	40 mm (9 h)	spring irrigation
Cycle 2023–2024					
Previous crop	SD/TC	2022–2023	chickpea	–	cereal–legume rotation
Deep tillage	TC	10 Oct 2023	three-share plough	–	primary soil preparation
Basal fertilisation	SD/TC	24 Dec 2023	matrix NP (7–17%)	150 kg·ha ⁻¹	basal NP supply
Sowing	TC	03 Jan 2024	conventional seeder	150 kg·ha ⁻¹ (TGW: 37 g)	density: 300 plants per m ²
Sowing	SD	03 Jan 2024	John Shearer no-till seeder	150 kg·ha ⁻¹ (TGW: 37 g)	density: 300 plants per m ²

cont. Tab. 1

Operation	System	Date	Material/Product	Rate/Parameter	Objective/Remarks
Chemical weeding 1	SD/TC	02 Jan 2024	glyphosate	4.5 dm ³ ·ha ⁻¹	non-selective; applied pre-planting to avoid crop damage
Chemical weeding 2	SD/TC	30 Jan 2024 and 22 Feb 2024	Floramix / Olympus Flex	320 g·ha ⁻¹ and 300 g·ha ⁻¹	selective post-emergence with safener; emergence/tillering (approved timings)
Nitrogen fertilisation 1	SD/TC	21 Feb 2024	urea (46 % N)	100 kg·ha ⁻¹	first supplementary application
Nitrogen fertilisation 2	SD/TC	25 Mar 2024	urea (46 % N)	100 kg·ha ⁻¹	second supplementary application
Irrigation 1	SD/TC	05 Feb 2024	sprinkler system	30 mm (6 h)	winter irrigation
Irrigation 2	SD/TC	13 Feb 2024	sprinkler system	40 mm (9 h)	spring irrigation
Irrigation 3	SD/TC	06 Mar 2024	sprinkler system	40 mm (9 h)	spring irrigation

Explanation: TGW = thousand grain weight (g)¹.

Source: own elaboration.

Stokes' law, with the results expressed as a percentage of fine earth (<2 mm).

Potential of hydrogen (pH) was measured in a soil-water suspension (1:2.5) using a pH meter (Hanna Instruments, Netherlands). Salinity was assessed by measuring the electrical conductivity (EC) of a saturated paste extract (Richards, 1954), reflecting the soluble salt content. Bulk density was determined using the cylinder method (Blake and Hartge, 1986).

The OC was quantified using the modified Walkley and Black method, employing potassium dichromate (K₂Cr₂O₇) oxidation in a sulphuric acid medium (Gillman, Sinclair and Beech, 1986). Total and active limestone (Ca_T and Ca_A) contents were obtained using a Bernard calcimeter (HCl-CaCO₃ reaction) (Sparks *et al.*, 2020) and the Drouineau method (Drouineau, 1942) (KMnO₄ titration), respectively. Total nitrogen (TN) was analysed using the Kjeldahl method, as described by Bremner (1965), which involves acid digestion and subsequent titration. Available phosphorus (P₂O₅) was extracted with the Olsen solution (0.5 M NaHCO₃, pH 8.5) and quantified by UV/visible spectrophotometry (Olsen *et al.*, 1954). Exchangeable potassium (K₂O) was measured by flame spectrometry after extraction with ammonium acetate (Nash, 1971), using a system equipped with a nebuliser, burner, and optical filter for atomic excitation. The cation exchange capacity (CEC) was determined using the sodium acetate method as described by Rhoades (1982).

ORGANIC CARBON STOCKS, SATURATION, DEFICIT, AND CRITICAL THRESHOLD

The soil organic carbon stock (SOCS) (Mg C·ha⁻¹) for the 0–30 cm depth was calculated for each sample using Equation (1) (Lefèvre *et al.*, 2017; FAO, 2019):

¹ Acc. to International System of Units, it should be mass of 1000 grains, however “thousand grain weight (TGW)” is the internationally established agronomic term widely used in seed science and cereal research.

$$SOCS = 0.1C \cdot BD \cdot T(1 - CP) \quad (1)$$

where: C = the carbon content (g C·kg⁻¹ soil), BD = the bulk density (g·cm⁻³), T = the thickness of the soil horizon (cm), and CP = the coarse particles (g·g⁻¹ soil).

The percentage of fine particles was used to calculate the stable carbon potential in soils (C_{sat}) using Equation (2) according to Hassink (1997):

$$C_{sat} = 4.09 + 0.37(\%cl + \%fsi) \quad (2)$$

where: cl = clay, fsi = fine silt.

The stable C value can be compared to the measured soil C and the carbon saturation deficit (C_{sat def}) can then be calculated (Angers *et al.*, 2011):

$$C_{sat def} = C_{sat} - C_{act} \quad (3)$$

where: C_{act} = actual carbon content.

The critical carbon threshold (C_{crit}) represents a minimum level of organic C below which the soil structure is weakened and the risk of degradation increases because of the low structural stability. This value was calculated according to Autfray *et al.* (2009) as follows:

$$C_{crit}(\%) = [0.32(\%cl + \%fsi) + 0.87]/10 \quad (4)$$

DESCRIPTIVE STATISTICS

A statistical analysis was conducted to summarise and compare the data from the two tillage systems: no-till (SD) and conventional tillage (TC). Descriptive statistics (mean, standard deviation, median, minimum, and maximum) were calculated for each measured parameter.

To evaluate the effect of the tillage system on the studied variables, a statistical analysis was performed on the data collected during the two growing seasons (2022 and 2024). First, the normality of the distribution of the quantitative variables (K₂O, P₂O₅, OC, N, CEC, C/N, Ca_A, pH, EC) was verified using the

Shapiro-Wilk test. Based on these results, comparisons of means between the two groups (SD vs TC) for each parameter were carried out using a Student's *t*-test for independent samples when the data followed a normal distribution, and the non-parametric Mann-Whitney U test when they did not. The significance level was set at $\alpha = 0.05$. A one-way analysis of variance (single classification ANOVA) was performed to assess the effect of the tillage system on the OC content.

All analyses were performed using the Python programming language (v.3.10), utilising the Pandas, SciPy.stats, and Statsmodels libraries.

Given the relatively short monitoring period (two growing seasons) and the moderate sample size ($n = 30$ per year, across both tillage systems), the statistical power to detect treatment effects may be limited. In addition, the observed within-plot variability may mask subtle differences between tillage systems. Consequently, non-significant results should be interpreted with caution, and the statistical analyses are intended to provide an initial assessment rather than definitive evidence of treatment effects.

PRINCIPAL COMPONENT ANALYSIS

To explore the multidimensional structure of the data and visualise the similarities and differences between the two tillage systems (SD and TC), a PCA was performed using the standard method described by Jolliffe and Cadima (2016).

This method reduced the data's dimensionality while retaining the majority of the total variance, thus facilitating a graphical interpretation of the relationships between variables and treatments. The PCA was carried out using the scikit-learn library in Python (v.3.10), supplemented by the Pandas, NumPy, and Matplotlib libraries for data processing and visualisation.

Because PCA is an exploratory multivariate technique, the observed graphical separations should be interpreted as indicative patterns and not as formal statistical proof of treatment differences.

PREDICTION METHODOLOGY

The future evolution of the SOC content was estimated through a linear extrapolation of the data measured between 2022 and 2024 for both cropping systems (SD and TC). A simple linear

regression model ($y = ax + b$) was applied to each time series, with the trend slopes calculated and the model validity verified using the coefficient of determination (R^2) and residual analysis.

This linear extrapolation approach has previously been used in the literature to provide initial estimates of SOC trends from limited time series data (Riggers *et al.*, 2021; Broeg *et al.*, 2024). While models such as RothC (Seitz *et al.*, 2025; Zhang *et al.*, 2025) are recognised as more robust tools for long-term projections, their application requires calibration with longer time series that are not yet available. As discussed by Addiscott (1993) and later by Smith *et al.* (2005), linear extrapolation can serve as a valid heuristic tool in this context. Our approach was therefore selected to provide an initial quantification of potential trajectories.

The trends were projected up to 2034, assuming stable agricultural practices and the absence of extreme climatic events. It is important to note that these assumptions represent a simplification, and that the projections should be interpreted as a comparison of the relative potentials of the systems rather than as absolute predictions (Paustian *et al.*, 2016; Paustian *et al.*, 2019). Nevertheless, this approach provides a clear and direct comparison of the carbon sequestration potential between systems over a timescale relevant to decision-makers. The analysis was performed using Python (v.3.10), employing the scipy.stats, linregress library for regressions and Matplotlib, Seaborn, and NumPy for visualisation and computation.

However, given the short duration of the observed time series, the extrapolated trajectories involve substantial uncertainty and should be regarded as heuristic scenarios rather than predictive forecasts.

RESULTS AND DISCUSSION

SOIL ORGANIC CARBON VARIATION ACROSS WHEAT PHENOLOGICAL STAGES

The dynamics of soil organic carbon (SOC) illustrated in Figure 2 reveal contrasting profiles between the SD and TC systems across the wheat growth stages (P1–P5). Figure 2 provides a descriptive visualisation of temporal patterns and does not, by itself, demonstrate statistically significant treatment effects. The SD system showed a visual tendency towards greater temporal stability, which may be related to improved retention of plant

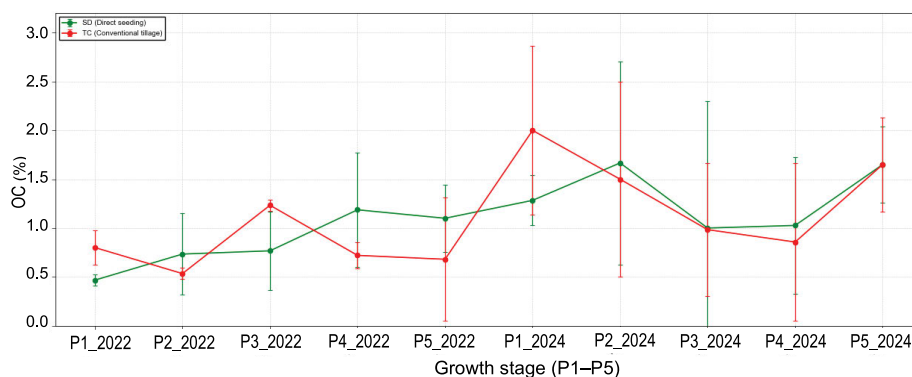


Fig. 2. Intra-plot variation of organic carbon (OC) under direct seeding (SD) and conventional tillage (TC) at different crop growth stages using combined data from 2022 and 2024; P1 = before sowing (06 Oct 2021 for 2021/2022, 06 Nov 2023 for 2023/2024), P2 = germination/emergence (08 Jan 2022, 31 Jan 2024), P3 = tillering (22 Feb 2022, 21 Feb 2024), P4 = heading stage (29 Mar 2022, 16 Apr 2024), P5 = physiological maturity (16 May 2022, 13 May 2024); points represent mean values; source: own study

residues and more consistent microbial activity in the absence of soil disturbance (Six *et al.*, 2002).

Given the short monitoring period and within-plot variability, the observed differences between treatments should be interpreted cautiously and considered indicative rather than conclusive.

Conversely, TC showed greater temporal variability, with temporary decreases at stages P2 and P4. These fluctuations may coincide with periods of tillage, which can accelerate OC mineralisation and increase spatial heterogeneity (West and Post, 2002). Descriptive statistical analysis supports this interpretation: the high and comparable standard deviations for OC in 2024 (0.80 for SD vs 0.76 for TC, Tab. 2) indicate substantial intra-plot variability in both systems, although the underlying mechanisms may differ. In addition, given the moderate sample size ($n = 30$ per year, across both tillage systems) and the variability observed within treatments, the statistical tests applied (t -test and ANOVA, $\alpha = 0.05$) may have limited power to detect subtle differences between systems.

The interannual comparison indicates an overall increase in OC content between 2022 and 2024 in both systems. OC increased from 0.79% to 1.33% under SD (+68% relative) and from 0.70% to 1.40% under TC (+100% relative). This general rise, together with the marked increase in CEC and the decrease in pH (Tab. 2), suggests that important changes occurred in the soil system between the two sampling campaigns. However, these interannual changes may also reflect climatic variability or seasonal effects and cannot be attributed solely to soil management.

Although the temporal patterns in Figure 2 suggest greater stability under SD, the mean values reported in Table 2 show no statistically significant difference in OC content between the two systems in 2024. Therefore, within the limits of the present dataset, the potential advantage of SD appears to relate more to the consistency of soil functioning and reduced disturbance than to a clear short-term increase in OC stocks. This interpretation remains consistent with the principles of conservation agriculture (Govaerts *et al.*, 2009; Derpsch *et al.*, 2010), but should be interpreted with caution.

STABILISATION AND CARBON LOSS PERIODS

The statistical analysis did not identify any significant differences in average OC content between tillage systems. The Student's t -test did not detect a statistically significant difference between TC and SD in 2022 ($t = 0.86$, $p = 0.39$) and again in 2024 ($t = 0.25$, $p = 0.80$). Similarly, ANOVA revealed no statistically significant effect of wheat phenological stages, regardless of tillage method or study year ($p \geq 0.05$). Overall, the F values suggest that the observed variations are largely attributable to inherent soil variability rather than to a strong effect of growth stages or cultivation technique. Given the relatively short monitoring period (two growing seasons) and the moderate sample size, the statistical power to detect treatment effects may have been limited.

The evolution of variability between years provides additional context. The SD tended to become more stable (2022: $F = 2.32$, $p = 0.12$; 2024: $F = 0.45$, $p = 0.76$), whereas TC showed a slight increase in variability (2022: $F = 0.57$, $p = 0.68$; 2024: $F = 1.09$, $p = 0.41$). Although no significant effect was detected, this pattern is consistent with the dominant role of natural spatial heterogeneity reported in cereal systems (Bambrick *et al.*, 2010).

However, examination of mean values by growth stage and tillage system reveals numerical tendencies that may reflect underlying biological and agronomic processes. As illustrated in Figure 3a, in 2022 SOC content ranged between 0.4% and 1.2%, remaining below commonly reported soil quality reference thresholds (1.4% for SD and 1.3% for TC). This situation may indicate increased vulnerability to structural and biological degradation, as values below these levels have been associated with reduced aggregate stability and microbial activity (Loveland and Webb, 2003). Lower values observed under TC are consistent with the enhanced mineralisation often reported following soil disturbance (Powlson *et al.*, 2014). In contrast, the slightly higher and more stable values observed under no-till may suggest improved carbon preservation associated with reduced mechanical disturbance, in agreement with trends reported in conservation agriculture studies (Bhattacharyya *et al.*, 2015).

Table 2. Descriptive statistics (mean $\pm$ standard deviation) of soil physicochemical properties under SD and TC (2021/2022 and 2023/2024)

Parameter	Statistics for			
	SD 2021/2022	TC 2021/2022	SD 2023/2024	TC 2023/2024
K ₂ O (g·kg ⁻¹)	0.27 $\pm$ 0.07	0.17 $\pm$ 0.08	0.11 $\pm$ 0.06	0.12 $\pm$ 0.06
N (%)	0.36 $\pm$ 0.08	0.40 $\pm$ 0.11	0.20 $\pm$ 0.03	0.20 $\pm$ 0.04
CEC (cmol·kg ⁻¹)	14.51 $\pm$ 1.85	14.08 $\pm$ 1.94	36.90 $\pm$ 12.52	36.83 $\pm$ 9.74
OC (%)	0.79 $\pm$ 0.35	0.70 $\pm$ 0.19	1.33 $\pm$ 0.80	1.40 $\pm$ 0.76
Rapport C/N	5.48 $\pm$ 4.07	4.93 $\pm$ 4.67	6.70 $\pm$ 4.17	7.27 $\pm$ 4.71
pH	7.67 $\pm$ 0.12	7.66 $\pm$ 0.15	7.38 $\pm$ 0.43	7.31 $\pm$ 0.46
P ₂ O ₅ (g·kg ⁻¹)	0.26 $\pm$ 0.11	0.33 $\pm$ 0.33	0.03 $\pm$ 0.02	0.04 $\pm$ 0.02
Ca _A (%)	5.78 $\pm$ 1.38	5.53 $\pm$ 1.02	4.84 $\pm$ 0.81	4.86 $\pm$ 0.46
EC (mS·cm ⁻¹)	0.17 $\pm$ 0.09	0.16 $\pm$ 0.09	0.82 $\pm$ 1.25	0.74 $\pm$ 1.21

Explanations: SD = direct seeding; TC = conventional tillage, CEC = cation exchange capacity, OC = organic carbon, Ca_A = active limestone, EC = electrical conductivity.

Source: own study.

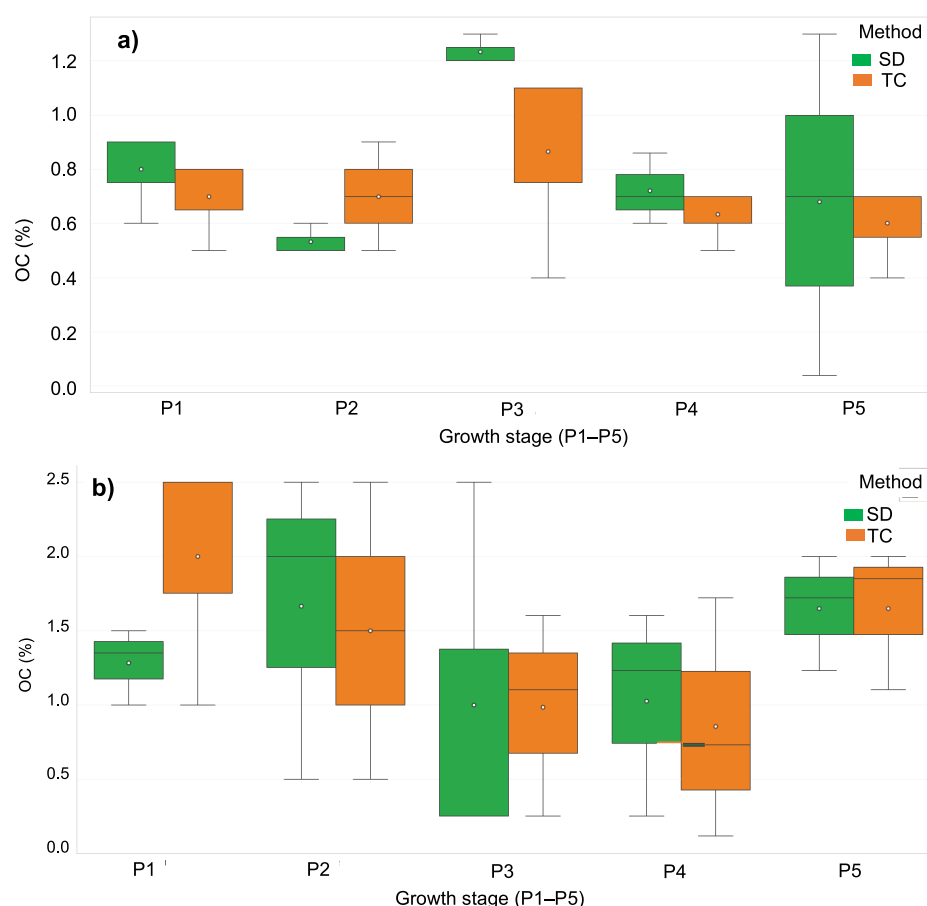


Fig. 3. Soil organic carbon variation according to wheat growth stages under direct seeding (SD) and conventional tillage (TC) in: a) 2022, b) 2024; P1–P5 as in Fig. 2; source: own study.

The comparison with Figure 3b (2024) indicates an overall increase in OC levels, now ranging between 1.0% and 2.5%. Under SD, mean values remained above the 1.4% reference level across stages, whereas TC approached but did not consistently exceed the 1.3% level at all stages. While these differences were not statistically significant, the observed pattern is broadly consistent with meta-analytical evidence reporting greater SOC accumulation under reduced tillage in certain environments (Álvaro-Fuentes *et al.*, 2008; Cui *et al.*, 2024). Visual inspection of the boxplots also suggests a modest seasonal dynamic, with a slight decline at tillering (P3) followed by recovery at later stages (P4–P5), possibly linked to root inputs and microbial activity as described in wheat systems (Quigley and Kravchenko, 2022).

Therefore, although no statistically significant differences were detected between tillage systems within the present dataset, the graphical trends suggest that SD may contribute to maintaining soil organic quality and resilience over time. This interpretation should be considered cautiously and confirmed through longer-term monitoring. Continuous tracking of OC by phenological stage remains relevant for the early detection of declines relative to the reference thresholds (1.4% for SD and 1.3% for TC), particularly under increasing climate variability (Lal, 2012).

EVOLUTION OF CHEMICAL FERTILITY PARAMETERS

The analysis of major nutrients between 2022 and 2024 highlights contrasting dynamics between the two tillage systems (Tab. 2). Potassium (K_2O) levels decreased markedly in both systems:

initially higher under SD ($0.27 \text{ g}\cdot\text{kg}^{-1}$) than under TC ($0.17 \text{ g}\cdot\text{kg}^{-1}$), they converged in 2024 toward similarly low values ($\approx 0.11\text{--}0.12 \text{ g}\cdot\text{kg}^{-1}$). The Student's *t*-test detected a statistically significant difference between systems in 2022 ($t = 3.30$; $p = 0.002$), with higher values under SD, whereas no significant difference was detected in 2024 according to the Mann–Whitney test ($U = 125.00$, $p = 0.61$). This overall decline may reflect progressive crop exportation, as reported in intensive cereal systems (Malhi, Kaur and Kaushik, 2021). Given the short duration of the study (two growing seasons), these nutrient dynamics should be interpreted as preliminary trends.

A marked decrease in P_2O_5 was also observed under both SD (0.26 to $0.03 \text{ g}\cdot\text{kg}^{-1}$) and TC (0.33 to $0.04 \text{ g}\cdot\text{kg}^{-1}$). This reduction may be related to mineral fixation processes (Hinsinger *et al.*, 2011), enhanced plant uptake stimulated by rhizosphere activity (Richardson and Simpson, 2011), or fertilisation insufficient to compensate for exports. However, the absence of statistically significant differences between systems in 2022 ($U = 113.50$, $p = 0.98$) and 2024 ($U = 123.50$, $p = 0.66$) calls for cautious interpretation. Nevertheless, the observed downward trend is consistent with the risk of progressive depletion of mobile nutrients reported in conservation systems (Vanlauwe *et al.*, 2015).

Concentrations of TN declined during the study period, stabilising at approximately 0.20% in 2024. The C/N ratio showed only a slight increase, with no significant differences between systems (SD: 6.70; TC: 7.27). Mann–Whitney tests did not detect statistically significant differences in N in 2022 ($U = 114.50$; $p = 0.95$) and 2024 ($p = 0.70$), and for the C/N ratio in 2022

($U = 110.00$, $p = 0.93$) and 2024 ($p = 0.72$). The relatively high intra-plot variability (standard deviations ≈ 4 –5) likely contributed to the lack of statistical discrimination and suggests the need for longer-term monitoring.

For cation exchange capacity (CEC), no statistically significant difference between tillage systems was detected ($p \geq 0.05$). Nevertheless, mean CEC values increased from $14.5 \text{ cmol}\cdot\text{kg}^{-1}$ in 2022 to $37 \text{ cmol}\cdot\text{kg}^{-1}$ in 2024. This increase coincided with the rise in OC and may reflect an overall improvement in soil chemical status, in line with the relationships described by Parikh and James (2012). The intermediate 2022/2023 season, during which the plots were under worked fallow with chickpea cultivation, may have contributed through the addition of labile organic residues and stimulation of microbial activity (Jensen, Peoples and Hauggaard-Nielsen, 2010; Jensen, Carlsson and Hauggaard-Nielsen, 2020). However, the high spatial variability (standard deviations $> 9 \text{ cmol}\cdot\text{kg}^{-1}$) indicates marked heterogeneity across plots and warrants cautious interpretation of the magnitude of change.

Soil pH showed a slight decrease in both systems (from 7.66 to 7.35 on average), possibly related to organic matter (OM) mineralisation processes (Curtin *et al.*, 2020). No significant difference between tillage systems was detected in 2022 ($t = 0.52$, $p = 0.60$) or 2024 ($U = 75.50$, $p = 0.12$), indicating a similar trend under SD and TC. Such changes may influence nutrient bioavailability (Kunhikrishnan *et al.*, 2016).

Finally, electrical conductivity (EC) increased slightly between 2022 and 2024 (≈ 0.7 – $0.8 \text{ mS}\cdot\text{cm}^{-1}$) but remained within agronomically acceptable limits. Although the Mann–Whitney test in 2024 indicated a statistically significant difference between systems ($U = 64.50$, $p = 0.04$), the small magnitude of the difference and the very close mean values suggest limited agronomic relevance. This isolated significance may be associated with gradual salt inputs from irrigation water ($3.15 \text{ mS}\cdot\text{cm}^{-1}$), applied twice in 2022 and three times in 2024.

TEMPORAL DYNAMICS OF SOIL ORGANIC CARBON

Soil organic carbon levels increased between 2022 and 2024 in both systems, rising from approximately 0.7–0.8% to 1.3–1.4%. Although the final mean content was slightly higher under TC,

the difference between systems remained small and was not statistically significant. Given the short monitoring period (two growing seasons), these temporal changes should be interpreted as preliminary trends rather than definitive treatment effects.

Analysis of SOCS also indicates this overall increase, with mean values approximately doubling in both systems. However, this increase was accompanied by greater intra-plot variability, as indicated by the high standard deviations observed in 2024, particularly under TC ($\approx 21 \text{ Mg}\cdot\text{ha}^{-1}$) compared with SD ($13.3 \text{ Mg}\cdot\text{ha}^{-1}$). This marked dispersion under TC suggests increased spatial heterogeneity and may be partly related to the disturbance effect of tillage on OM distribution, as reported by Six *et al.* (2002). In contrast, the slightly lower variability observed under SD may indicate relatively greater temporal stability over the study period. Moreover, considering the moderate sample size and the variability observed within treatments, the statistical power of the applied tests may have been insufficient to detect subtle differences between tillage systems.

The interannual trend also shows a concurrent increase in SOC and CEC between 2022 and 2024 in both systems. This period followed the 2022/2023 season, during which the plots were under worked fallow with chickpea cultivation. As a legume, chickpea is known to contribute organic residues and enhance soil biological activity (Jensen, Peoples and Hauggaard-Nielsen, 2010; Carlsson and Hauggaard-Nielsen, 2020), which may have supported the observed improvement. Nevertheless, given the absence of significant differences between tillage systems, the increase is more likely attributable to the cereal–legume rotation effect common to both treatments rather than to tillage alone.

PRINCIPAL COMPONENT ANALYSIS OF SOIL PROPERTIES: SOIL MANAGEMENT SYSTEM DIFFERENTIATION BETWEEN 2022 AND 2024

The PCA was used as an exploratory multivariate tool and was complemented by univariate statistical tests (ANOVA and Student's *t*-test, $p < 0.05$) applied to individual soil variables. The two PCA analyses performed on data from 2022 (Fig. 4a) and 2024 (Fig. 4b) allow examination of changes in soils subjected to two contrasting management methods: SD and TC. Combining

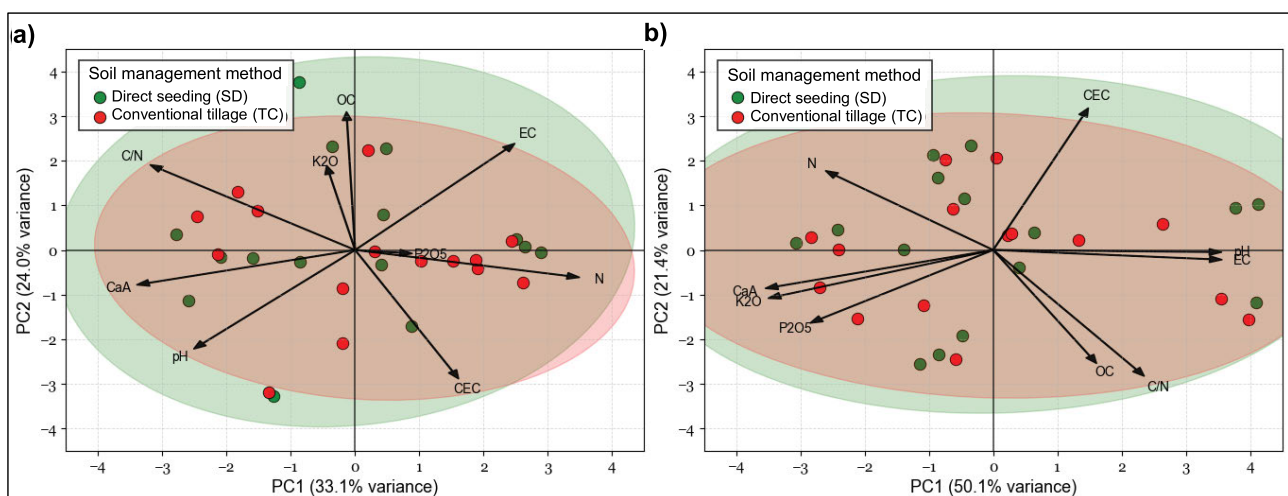


Fig. 4. Principal component analysis comparison of soil properties under conventional tillage (TC) and direct seeding (SD) in: a) 2022, b) 2024; ellipses represent 95% confidence regions and are shown for descriptive purposes only; source: own study

these two analyses provides insight into both the immediate effects of cultivation practices and their temporal dynamics.

In 2022, the first two principal components accounted for 57.1% of the total variance (PC1 = 33.1%; PC2 = 24.0%), while in 2024 this cumulative variance increased to 71.5% (PC1 = 50.1%; PC2 = 21.4%). This increase suggests a somewhat clearer organisation of the multivariate structure in 2024; however, the substantial overlap between treatments indicates that differences between agricultural practices remained limited within the short-term observation period.

The PCA reveals contrasting patterns between the two systems. In 2022, TC appears to be associated with higher values of N, CEC, P_2O_5 , and EC, which may reflect processes such as accelerated mineralisation and more intense chemical activity, in line with the disturbance mechanisms described by Alvarez and Steinbach (2009) and Cárcer *et al.* (2019). In 2024, these patterns persist, accompanied by increased sample dispersion, suggesting spatial heterogeneity induced by repeated ploughing. This phenomenon, documented in Mediterranean cereal systems by Nunes, Karlen and Moorman (2020), may have implications for soil stability. Additionally, TC remains associated with indicators of increased salinity and alkalinity, conditions that could potentially affect long-term nutrient availability.

Conversely, in 2022, SD showed a tendency toward association with higher OC content, a higher C/N ratio, and elevated concentrations of K_2O and Ca_A , suggesting better preservation of organic matter, as observed by Sá *et al.* (2024) in conservation systems. It should be noted, however, that these patterns represent trends observed in the multivariate space and should be interpreted with caution in the absence of statistical validation for each individual parameter.

In 2024, this tendency appeared to strengthen, with a visual pattern of clearer structuring of samples around the OC and C/N variables. These graphical patterns are consistent with the hypothesis that SD may gradually promote improvement in the biological and chemical quality of the soil, potentially optimising organic matter stability through the physical protection mechanisms of aggregates (Blanco-Canqui and Ruis, 2018) and nutrient availability, particularly in surface horizons (Corbeels *et al.*, 2019). The PCA ordination suggests a tendency toward differentiated responses of soil trajectories to management practices, though further statistical validation would be needed to confirm these patterns.

The comparative PCA analysis reveals evolving patterns between management systems – TC may be consistent with processes such as accelerated mineralisation but shows trends suggesting a gradual deterioration in soil chemical consistency, with marked spatial heterogeneity. Meanwhile, SD exhibits graphical patterns consistent with cumulative benefits, suggesting potential improvement in soil structure, OC content, and nutrient storage (K_2O , Ca_A) after two growing seasons. These observations align with the conclusions of Pittelkow *et al.* (2015) on the temporal effects of conservation practices, though they should be regarded as indicative trends rather than statistically confirmed effects.

The differential evolution observed in the PCA suggests that farming practices may influence not only the immediate quality of soils but also their medium-term trajectory. The patterns observed are consistent with the potential of SD as a sustainable

strategy whose advantages may increase over time through improved organic stability and enhanced nutrient availability in surface horizons, as demonstrated by recent meta-analyses in semi-arid areas (Kassam, Friedrich and Derpsch, 2019). However, given the exploratory nature of this study and the short-term dataset, these interpretations should be considered as working hypotheses requiring confirmation through long-term monitoring and statistical validation.

The PCA biplots, complemented by 95% confidence ellipses, revealed a substantial overlap between SD and TC treatments in both years (Fig. 4). This pattern suggests that the differentiation between soil management systems remained limited within the short-term observation period. A slight tendency toward structuring along PC1 was observed in 2024; however, the broad overlap of the confidence regions indicates that the PCA results should be interpreted as exploratory rather than as evidence of clear multivariate separation. Furthermore, the moderate sample size may have limited the ability of multivariate analysis to detect subtle treatment effects.

PROJECTION OF ORGANIC CARBON CONTENTS

Linear projections for 2034 (Fig. 5) suggest differences between the two tillage systems – SD shows an upward numerical trend, with a projected OC content of 2.3% in 2034, compared with 1.2% for TC. These projections remain below theoretical saturation thresholds (19% for SD and 17% for TC). While these trends may indicate potential for enhanced soil carbon under SD, it should be noted that statistical significance for these long-term differences has not been established.

By 2024, SD is projected to exceed the critical threshold of 1.4%, recognised as necessary for maintaining structural stability and soil fertility (Autfray *et al.*, 2009; Angers *et al.*, 2011), while TC is projected to remain below this threshold. These projections highlight possible differences in carbon dynamics associated with tillage practices, but should be interpreted cautiously, particularly given the semi-arid conditions that naturally favour mineralisation (Paustian *et al.*, 2016; Lal, 2020).

The 95% confidence intervals are broader under SD, reflecting higher system responsiveness rather than guaranteed increases, while the lower variability projected for TC indicates more limited responsiveness.

The substantial gap between projected levels and theoretical saturation limits suggests considerable potential for further SOC accumulation. As demonstrated across various agroecosystems – including grasslands, Mediterranean agrosilvopastoral systems, and agroforestry – soils may not have reached their carbon saturation limit and thus hold great potential for further SOC sequestration (Shi *et al.*, 2018; Bai and Cotrufo, 2022; Alvarez-Castellanos *et al.*, 2024). Further SOC accumulation could be achieved through appropriate management strategies, including: (i) optimising organic inputs via legume integration, residue management, manure application, or cover crops to enhance microbial necromass contribution to stable carbon pools (Bai and Cotrufo, 2022; Alvarez-Castellanos *et al.*, 2024); (ii) mitigating carbon losses during warm periods by maintaining soil cover or modifying microclimate (e.g., through shading or increased soil moisture) to reduce mineralisation rates (Shi *et al.*, 2018; Bai and Cotrufo, 2022); and (iii) promoting vertical carbon distribution beyond surface horizons through deep-rooting crops or agroeco-

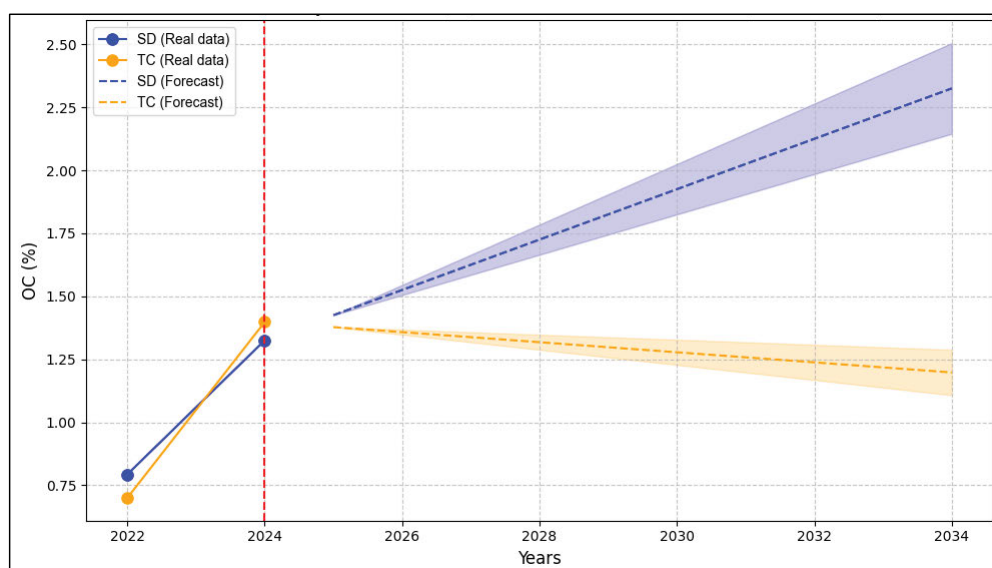


Fig. 5. Projected soil organic carbon trends until 2034 under two tillage systems: direct seeding (SD) and conventional tillage (TC); the shaded area represents the 95% confidence interval, and the vertical red line indicates the end of the observed data period; projections are based on short-term linear trends and should be interpreted as indicative rather than predictive; source: own study

logical practices that increase root-derived carbon inputs to subsoil layers, where mean residence time is three to ten times longer than in topsoil (Baker *et al.*, 2007; Cotrufo *et al.*, 2013; Shi *et al.*, 2018; Bai and Cotrufo, 2022; Alvarez-Castellanos *et al.*, 2024).

Given the heuristic nature of linear extrapolation, these results should not be interpreted as precise forecasts. They provide a preliminary indication of trends, which could inform further modelling using tools such as RothC (Coleman and Jenkinson, 1996) or Century (Parton *et al.*, 1987). Continuous monitoring over longer periods (5–10 years) is essential to obtain robust time series data, enabling a transition from trend projections to reliable predictions and a more accurate assessment of carbon resilience under SD in semi-arid contexts. It should be noted that the linear projections assume constant management practices and do not account for interannual climate variability, which may influence SOC dynamics. The statistical significance tests applied to observed data (*t*-test, Mann-Whitney *U*) cannot be extrapolated to these projections, and their validity is limited by the short duration of the time series. Consequently, these results provide indicative trends rather than statistically confirmed forecasts, and caution is advised when interpreting long-term differences between SD and TC.

COMPARATIVE IMPACTS OF SOIL MANAGEMENT SYSTEMS

The results of this study indicate some differences in dynamics between SD and TC. By retaining residues on the surface, SD may provide physical protection of the soil, which can help stabilise microbial activity (Pittelkow *et al.*, 2015), preserve aggregate structure (Blanco-Canqui and Ruis, 2018), and contribute to gradual accumulation of OM in surface horizons (Corbeels *et al.*, 2019). Over our two-year monitoring period, numerical trends suggested an increase in OC under SD, but these differences were not statistically tested for long-term significance and should therefore be interpreted with caution.

Conversely, TC, through repeated mechanical disturbance, can expose OM to increased mineralisation (Alvarez and Steinbach, 2009) and disrupt aggregates (Nunes, Karlen and Moorman, 2020). Over the course of this study, the results showed a numerical decrease in OC under TC and increased dispersion of sample values in 2024, which may indicate growing heterogeneity potentially linked to differential tillage impacts on soil properties (Sá *et al.*, 2024).

The observed divergence between these systems suggests potential benefits of SD for soil carbon maintenance, particularly in semi-arid environments (Kassam, Friedrich and Derpsch, 2019). Projections for 2034, however, are heuristic and based on linear extrapolation of short-term data, and should not be interpreted as predictive. These projections indicate possible continued carbon accumulation under SD and a relative decline under TC, but statistical support for these long-term trends is lacking.

Further research integrating biological indicators and long-term monitoring would strengthen the understanding of soil carbon responses. While our study provides evidence of trends consistent with the potential advantages of SD, conclusions regarding its long-term effectiveness for soil fertility and climate mitigation should remain cautious and framed as indicative rather than definitive (Smith *et al.*, 2021; Pretty and Bharucha, 2014).

CONCLUSIONS

This study suggests that direct seeding may contribute to improved soil organic carbon dynamics under semi-arid conditions. The principal component analysis and the ten-year projection indicate a potential trend toward enhanced soil carbon stability under direct seeding, although statistical significance was not consistently observed.

The results further suggest that the potential advantage of direct seeding lies less in short-term increases in soil organic

carbon than in its capacity to improve the temporal and spatial stability of soil carbon, thereby promoting a more resilient sequestration trajectory.

However, these findings are based on a short monitoring period and should be interpreted with caution until confirmed by long-term studies.

Thus, direct seeding may represent a promising alternative for semi-arid cereal systems, potentially contributing to both agricultural sustainability and climate change mitigation. Its widespread adoption, supported by appropriate policies, could contribute to food security and soil conservation in Algeria and other Mediterranean regions.

ABBREVIATIONS

BD = bulk density ($\text{g}\cdot\text{cm}^{-3}$)
 C_{act} = actual carbon (%)
 C_{crit} = critical thresholds of carbon (%)
 C_{sat} = carbon saturation (%)
 $C_{\text{sat def}}$ = carbon saturation deficit (%)
 cl = clay (%)
 fsi = fine silt (%)
 C/N = carbon to nitrogen ratio
 Ca_A = active limestone (%)
 Ca_T = total limestone (%)
 CEC = cation exchange capacity ($\text{cmol}\cdot\text{kg}^{-1}$)
 CP = gravel content ($\text{g}\cdot\text{g}^{-1}$)
 EC = electrical conductivity ($\text{mS}\cdot\text{cm}^{-1}$)
 TN = total nitrogen (%)
 NP = nitrogen-phosphorus fertiliser
 OC = organic carbon (%)
 OM = organic matter (%)
 pH = potential of hydrogen (–)
 PCA = principal component analysis
 R^2 = coefficient of determination (–)
 SD = direct seeding
 SOC = soil organic carbon
 $SOCS$ = soil organic carbon stocks ($\text{Mg C}\cdot\text{ha}^{-1}$)
 T = the thickness of the soil horizon (cm)
 TC = conventional tillage
 TGW = thousand grain weight, expressed as the mass of 1000 grains (g)

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CONFLICT OF INTERESTS

All authors declare that they have no conflicts of interest.

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